OE-EC804A: Artificial Intelligence

Quick-Recall One-Liners Revision Guide

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Chapter 1: Introduction to Artificial Intelligence

1. **Artificial Intelligence definition** → Branch of computer science building smart, autonomous systems.
2. **Standard modern approach to AI** → Acting Rationally (Rational Agent approach).
3. **Two dimensions of conceptual approaches to AI** → Thought processes vs. behavior; human-like vs. rational.
4. **Thinking Humanly focus** → Mimicking cognitive activities and mental processes of humans.
5. **Thinking Rationally focus** → Using formal logic and laws of thought.
6. **Acting Humanly focus** → Performing actions indistinguishable from a human.
7. **Acting Rationally focus** → Maximizing expected performance or payoff.
8. **Inventor of the Turing Test** → Alan Turing.
9. **Year Turing Test was introduced** → 1950.
10. **Original name of the Turing Test** → The Imitation Game.
11. **Duration of interrogation in Turing Test** → 5 minutes.
12. **NLP purpose in Turing Test** → To communicate successfully in natural language.
13. **Knowledge Representation purpose in Turing Test** → To store information acquired before or during conversation.
14. **Automated Reasoning purpose in Turing Test** → To draw conclusions from stored information.
15. **Machine Learning purpose in Turing Test** → To adapt to new circumstances and detect patterns.
16. **Two additional requirements for Total Turing Test** → Computer Vision and Robotics.
17. **Cognitive Science discipline contribution to AI** → Mimicking human cognitive processes.
18. **Philosophy contribution to AI** → Rules of logic and mind-machine concepts.
19. **Mathematics contribution to AI** → Logic, probability, algorithms, and tractability.
20. **Economics contribution to AI** → Utility theory, decision theory, and game theory.
21. **Neuroscience contribution to AI** → Biological models of information processing (ANNs).
22. **Psychology contribution to AI** → Behavioral and cognitive models of human action.
23. **Control Theory contribution to AI** → Feedback loops and self-regulating systems.
24. **Computer Vision main function** → Extracting structured information from digital images or videos.
25. **Natural Language Processing main function** → Processing and analyzing human languages.
26. **Machine Learning main function** → Improving task performance automatically from train-

ing data.

27. **Expert System definition** → Software emulating human expert decision-making in a narrow domain.
28. **Three main components of an Expert System** → Knowledge Base, Inference Engine, and User Interface.
29. **Expert System Knowledge Base storage format** → IF-THEN production rules.
30. **Forward Chaining search bias** → Data-driven reasoning from facts to conclusions.
31. **Backward Chaining search bias** → Goal-driven reasoning starting from hypothesis to facts.
32. **Explanation Facility purpose** → Showing the step-by-step reasoning path of conclusion.
33. **Major disadvantage of Expert Systems** → Lack of general common-sense knowledge.
34. **Knowledge Acquisition Bottleneck definition** → Difficulty of extracting tacit knowledge from human experts.
35. **System brittleness in Expert Systems** → Crashing due to gaps or contradictions in rules.
36. **Year of the Dartmouth workshop (AI birth)** → 1956.
37. **Who coined the term "Artificial Intelligence"?** → John McCarthy.

Chapter 2: Intelligent Agents

1. **AI Agent definition** → Anything perceiving environment through sensors and acting via actuators.
2. **Percept Sequence definition** → Complete history of all percepts received to date.
3. **Agent Function mathematical mapping** → $f : P^* \to A$
4. **Agent Program definition** → Concrete implementation of agent function on physical architecture.
5. **Agent formula** → Agent = Architecture + Program
6. **PEAS acronym components** → Performance, Environment, Actuators, Sensors.
7. **PEAS Performance Measure for Automated Taxi** → Safe arrival, passenger comfort, legal driving, and profit.
8. **PEAS Environment for Automated Taxi** → Roads, traffic, pedestrians, weather, and customers.
9. **PEAS Actuators for Automated Taxi** → Steering wheel, accelerator, brakes, horn, and display.
10. **PEAS Sensors for Automated Taxi** → Cameras, LIDAR, sonar, GPS, and speedometer.
11. **PEAS Sensors for Medical Diagnosis System** → Keyboard or voice input of symptoms.
12. **Autonomy in agents** → Behavior guided by own experience, not built-in designer knowledge.
13. **Five basic agent structures** → Simple reflex, model-based, goal-based, utility-based, learning agents.
14. **Simple Reflex Agent action driver** → Current percept only, using condition-action rules.
15. **Key limitation of Simple Reflex Agents** → Requires a fully observable environment.
16. **Model-Based Reflex Agent key feature** → Maintains internal state to track unobserved world aspects.
17. **Goal-Based Agent action selector** → Chooses actions that achieve explicit goal states.

18. **Utility-Based Agent action selector** → Maximizes a real-valued utility function indicating state preference.
19. **Four components of a Learning Agent** → Learning element, performance element, critic, problem generator.
20. **Learning Element function** → Making improvements by learning from experience.
21. **Performance Element function** → Selecting external actions to execute.
22. **Critic function** → Evaluating behavior against external performance standards.
23. **Problem Generator function** → Suggesting exploratory actions to discover new knowledge.
24. **Fully Observable environment** → Sensors detect complete state of environment at each point.
25. **Partially Observable environment** → Sensors have noise or gaps leaving state details unknown.
26. **Deterministic environment** → Next state completely determined by current state and action.
27. **Stochastic environment** → Next state contains random uncertainty.
28. **Strategic environment** → Uncertainty arises only from actions of other agents.
29. **Episodic environment** → Independent, atomic episodes where past actions don't affect future.
30. **Sequential environment** → Current decisions affect all future states and actions.
31. **Static environment** → Environment does not change during agent decision-making.
32. **Dynamic environment** → Environment changes while agent is deciding.
33. **Semidynamic environment** → Environment does not change, but performance score decreases with time.
34. **Discrete environment** → Finite or countable states, percepts, and actions.
35. **Continuous environment** → Continuous real variables represent environment states.
36. **Chess with clock environment properties** → Fully observable, strategic, sequential, semi-dynamic, discrete, multi-agent.
37. **Taxi driving environment properties** → Partially observable, stochastic, sequential, dynamic, continuous, multi-agent.

Chapter 3: Solving Problems by Searching (Uninformed Search)

1. **Problem-Solving Agent definition** → Goal-based agent finding action sequences leading to goal states.
2. **Five components of problem formulation** → Initial state, actions, transition model, goal test, path cost.
3. **Transition Model definition** → Function $Result(s, a)$ returning state reached by action.
4. **Path Cost formula** → $g(n)$ = sum of step costs along path.
5. **State Space representation** → Directed graph with states as nodes and actions as edges.
6. **Four search evaluation criteria** → Completeness, time complexity, space complexity, optimality.
7. **Branching factor (b)** → Maximum number of successors of any node.

8. **Uninformed Search definition** → Search using only problem definition without heuristic guidance.
9. **BFS search strategy** → Explores level-by-level, expanding all shallowest nodes first.
10. **BFS frontier data structure** → FIFO Queue (First-In, First-Out).
11. **BFS Space Complexity** → $O(b^d)$ (Exponential memory).
12. **BFS Optimality condition** → Step costs must be uniform.
13. **DFS search strategy** → Explores deep into a branch before backtracking.
14. **DFS frontier data structure** → LIFO Stack or Recursion.
15. **DFS Space Complexity** → $O(bm)$ (Linear memory).
16. **Reason DFS is not complete** → Can get trapped in infinite paths or loops.
17. **DLS definition** → DFS running with a pre-specified depth limit l.
18. **IDDFS definition** → Repeated DLS with increasing limits ($l = 0, 1, 2...$).
19. **IDDFS Space Complexity** → $O(bd)$ (Linear space).
20. **IDDFS Completeness status** → Complete (if b is finite).
21. **UCS node expansion criteria** → Node with lowest accumulated path cost $g(n)$.
22. **UCS frontier data structure** → Priority Queue.
23. **UCS Time and Space Complexity** → $O(b^{1+\lfloor C^*/\epsilon \rfloor})$.
24. **UCS completeness condition** → Step costs must be positive ($\geq \epsilon > 0$).
25. **8-Puzzle Initial State** → Scrambled configuration of 8 numbered tiles in 3×3 grid.
26. **8-Puzzle Actions** → Moving blank space Left, Right, Up, or Down.
27. **Missionaries and Cannibals safety constraint** → Cannibals must never outnumber missionaries on either bank.
28. **Missionaries and Cannibals State representation** → (M, C, B) for Left bank values.
29. **Goal state in River Crossing problem** → $(0, 0, 0)$.
30. **Water Jug Problem initial state** → $(0, 0)$ (both jugs empty).
31. **Water Jug Problem Goal State** → $(2, y)$ where 4-gallon jug has 2 gallons.
32. **Water Jug rule for filling 3-Gal jug** → $(x, 3)$ if $y < 3$.
33. **Water Jug rule for pouring 4-Gal into 3-Gal until full** → $(x - (3 - y), 3)$ if $x + y \geq 3$.
34. **Water Jug rule for pouring all 3-Gal into 4-Gal** → $(x + y, 0)$ if $x + y \leq 4$.

Chapter 4: Informed Search & Exploration

1. **Informed Search definition** → Search using problem-specific heuristic knowledge to guide pathfinding.
2. **Heuristic function $h(n)$** → Estimated cost of cheapest path from n to goal.
3. **Greedy Best-First Search evaluation function** → $f(n) = h(n)$.
4. **Greedy Best-First Search expansion bias** → Expands node closest to goal.
5. **Greedy Best-First Search space complexity** → $O(b^l)$ (Exponential space).
6. **A* Search evaluation function** → $f(n) = g(n) + h(n)$.
7. **A* Search completeness condition** → Branching factor finite; step costs $\geq \epsilon > 0$.

8. **Admissible heuristic definition** → Heuristic never overestimating true cost to goal ($0 \leq h(n) \leq h^*(n)$).
9. **Consistent (monotone) heuristic definition** → $h(n) \leq c(n, a, n') + h(n')$ (triangle inequality).
10. **A* tree search optimality condition** → Heuristic $h(n)$ must be admissible.
11. **A* graph search optimality condition** → Heuristic $h(n)$ must be consistent.
12. **Relation between consistency and admissibility** → Every consistent heuristic is admissible.
13. **Monotonicity of f-values in A*** → If $h(n)$ is consistent, $f(n)$ is non-decreasing along paths.
14. **Manhattan Distance heuristic formula for 8-Puzzle** → Sum of horizontal and vertical distances from goal.
15. **Misplaced Tiles heuristic definition** → Count of tiles not in their goal position.
16. **Simulated Annealing working principle** → Local search accepting bad moves with decreasing probability over time.
17. **Simulated Annealing acceptance probability formula** → $P = e^{\frac{\Delta E}{T}}$.
18. **Impact of slow cooling in Simulated Annealing** → Guaranteed to find global optimum with probability 1.
19. **Impact of fast cooling in Simulated Annealing** → Traps system in a local optimum.
20. **Hill-Climbing Search definition** → Greedy local search moving to neighbor with highest value.
21. **Three limitations of Hill-Climbing** → Local Maxima, Ridges, Plateaus.
22. **Flat local maximum vs. Shoulder on a Plateau** → Flat maximum has no exit; shoulder allows progression.
23. **Solution to Hill-Climbing limitations** → Random-Restart Hill Climbing.
24. **Local Beam Search working principle** → Tracks k states in parallel, keeping best k successors.
25. **AO* Search application** → Finding solutions in AND-OR graphs (decomposable problems).
26. **AO* evaluation function for AND branch** → $f(n) = g(n) + \sum h(\text{sub-problems})$.
27. **Recursive Best-First Search (RBFS) Space Complexity** → $O(bd)$ (Linear space).
28. **RBFS backtracking trigger** → When current node f-value exceeds the f-limit.
29. **RBFS back-propagation mechanism** → Replaces parent's f-value with best child's f-value on backtrack.
30. **RBFS time complexity limitation** → Exponential time due to node regeneration.

Chapter 5: Constraint Satisfaction Problems

1. **Factored state representation in CSPs** → States are represented by variables, domains, and constraints.
2. **Three core components of a CSP** → Variables (V), Domains (D), Constraints (C).
3. **Australia map-coloring variables** → $\{WA, NT, Q, NSW, V, SA, T\}$.
4. **Australia map-coloring domain values** → $\{\text{Red}, \text{Green}, \text{Blue}\}$.
5. **Australia map-coloring constraint type** → Adjacent regions must have different colors

$(WA \neq NT)$.

6. **Tasmania constraint isolation in Australia map** → Tasmania has no constraints with other territories.
7. **Cryptarithmetic variables in SEND + MORE = MONEY** → Letters $\{S, E, N, D, M, O, R, Y\}$ and carries $\{C_1, C_2, C_3, C_4\}$.
8. **Domain of leading letters (S, M) in Cryptarithmetic** → $\{1..9\}$ (cannot be zero).
9. **Alldifferent Constraint definition** → Requires all variables in a set to be unique.
10. **Column constraint for units column in SEND+MORE** → $D + E = Y + 10 \times C_1$.
11. **Solved value of M in SEND+MORE** → $M = 1$.
12. **Solved value of S in SEND+MORE** → $S = 9$.
13. **Solved value of O in SEND+MORE** → $O = 0$.
14. **Final Cryptarithmetic equation solution** → $9568 + 1085 = 10653$.
15. **Flexible CSP definition** → CSP where soft constraints/preferences are relaxed to resolve overconstraints.
16. **Goal of Flexible CSP solver** → Maximize preference utility or minimize violation penalty costs.
17. **Backtracking Search strategy** → Depth-first search instantiating variables sequentially, checking constraints.
18. **Backtracking search data structure basis** → LIFO stack and recursion.
19. **Standard logic language for Constraint Programming** → Prolog.
20. **Prolog library for finite domain constraint solving** → CLP(FD).
21. **Chronological backtracking limit** → Tends to repeat same failures down search tree (thrashing).
22. **Constraint propagation definition** → Using constraints to reduce variable domains before or during search.
23. **Node consistency definition** → Every value in variable's domain satisfies unary constraints.
24. **Arc consistency (AC-3) definition** → For every X value, some Y value is allowed.

Chapter 6: Adversarial Search

1. **Minimax definition** → Recursive decision algorithm for two-player, zero-sum, perfect-information games.
2. **Zero-sum game definition** → Opponent's gain is exactly equal to player's loss.
3. **Player MAX objective in Minimax** → Maximize the final utility score.
4. **Player MIN objective in Minimax** → Minimize the MAX utility score.
5. **Value propagated from child at a MAX node** → Maximum of the child values.
6. **Value propagated from child at a MIN node** → Minimum of the children's values.
7. **Minimax Time Complexity** → $O(b^d)$ (Exponential).
8. **Minimax Space Complexity** → $O(bd)$ (Linear space).
9. **Alpha-Beta Pruning definition** → Optimization skipping subtrees that won't change minimax decision.
10. **Alpha (α) in pruning** → Best (highest) choice found so far for MAX.

11. **Beta (β) in pruning** → Best (lowest) choice found so far for MIN.
12. **Initial values of Alpha and Beta** → $\alpha = -\infty$ and $\beta = +\infty$.
13. **Pruning cutoff condition** → When $\beta \leq \alpha$.
14. **Alpha-cutoff location** → Occurs at a MIN node when $\beta \leq \alpha$.
15. **Beta-cutoff location** → Occurs at a MAX node when $\beta \leq \alpha$.
16. **Time complexity of Alpha-Beta pruning with perfect ordering** → $O(b^{d/2})$ (doubles solvable depth).
17. **Does pruning change the optimal minimax decision?** → No, it returns the identical move.
18. **Leaf node evaluation order effect on pruning** → Better moves evaluated first increase pruned branches.
19. **Minimax game tree level representation** → Alternate plies of MAX moves and MIN moves.
20. **Evaluation function in real-time games** → Estimates state utility when search limit is reached.

Chapter 7: Logical Agents

1. **Propositional Logic connectives** → Negation ($\neg$), conjunction ($\wedge$), disjunction ($\vee$), implication ($\rightarrow$), biconditional ($\leftrightarrow$).
2. **Soundness of inference** → Algorithm derives only sentences that logically follow (derives truth).
3. **Completeness of inference** → Algorithm derives all sentences that logically follow.
4. **Tautology definition** → Sentence that is True under all possible models.
5. **Contradiction (unsatisfiable) definition** → Sentence that is False under all possible models.
6. **Contingency definition** → Sentence True in some models, False in others.
7. **Peirce's Law formula** → $(((P \rightarrow Q) \rightarrow P) \rightarrow P)$ (tautology).
8. **Modus Ponens syntax** → $\frac{P, \quad P \rightarrow Q}{Q}$.
9. **Modus Tollens syntax** → $\frac{\neg Q, \quad P \rightarrow Q}{\neg P}$.
10. **Resolution rule syntax** → $\frac{P \vee Q, \quad \neg P \vee R}{Q \vee R}$.
11. **Unit Resolution syntax** → $\frac{P \vee Q, \quad \neg P}{Q}$.
12. **Knowledge-Based Agent definition** → Agent utilizing structured Knowledge Base (KB) for logical reasoning.
13. **KB-Agent TELL function** → Adds new facts or percepts to the KB.
14. **KB-Agent ASK function** → Queries the KB for optimal next actions.
15. **PEAS Environment of Wumpus World** → 4×4 grid cave containing Wumpus, pits, and gold.
16. **Stench sensor trigger in Wumpus World** → Adjacent rooms to the Wumpus.
17. **Breeze sensor trigger in Wumpus World** → Adjacent rooms to pits.
18. **Glitter sensor trigger in Wumpus World** → Room containing the gold.
19. **Wumpus World performance penalty for death** → -1000 points.

20. **Semantics in propositional logic** → Rules computing truth value of sentences in a model.
21. **Standard proposition symbols in AI** → 2 constant symbols: True ($\top$) and False ($\perp$).
22. **Validity inference relation** → KB $\models \alpha$ if and only if (KB $\rightarrow \alpha$) is valid.
23. **Satisfiability inference relation** → KB $\models \alpha$ if and only if (KB $\wedge \neg\alpha$) is unsatisfiable.
24. **CNF sentence definition** → Conjunctive Normal Form (conjunction of disjunctions of literals).
25. **Original statement satisfiability under CNF conversion** → Satisfiability is preserved (CNF unsatisfiable implies original unsatisfiable).
26. **Unit Clause definition** → A clause containing exactly a single literal.
27. **Why Resolution is a single inference rule** → It is refutation-complete for reasoning.

Chapter 8: First-Order Logic & Knowledge Representation

1. **Declarative Knowledge definition** → Static facts and assertions representing concepts ("knowing what").
2. **Procedural Knowledge definition** → Steps, rules, and procedures to perform tasks ("knowing how").
3. **Production System components** → Production rules (Rule Base), Working Memory, and Inference Engine.
4. **Production System rule format** → IF $\langle$Condition$\rangle$ → THEN $\langle$Action$\rangle$.
5. **Recognize-Act cycle steps** → Match rules, resolve conflict, execute action.
6. **Horn Clause definition** → A clause containing at most one positive literal.
7. **Definite Clause definition** → Horn clause with exactly one positive literal.
8. **Implication conversion to Horn Clause ($p \rightarrow q$)** → $\neg p \vee q$ (definite clause).
9. **Fact in Horn Clauses** → Horn clause with one positive, zero negative literals.
10. **Goal Clause in Horn Clauses** → Horn clause with zero positive literals.
11. **Skolemisation definition** → Eliminating existential quantifiers ($\exists$) by substituting concrete terms.
12. **Skolem Constant substitution** → $\exists x P(x) \Longrightarrow P(A)$ (no universal scope).
13. **Skolem Function substitution** → $\forall x \exists y P(x,y) \Longrightarrow \forall x P(x, F(x))$ (universal scope).
14. **Prolog programming paradigm** → Declarative logic programming language based on Horn clauses.
15. **Prolog base case for factorial of 0** → ‘factorial(0, 1).‘
16. **Prolog mod operator in GCD recursive rule** → ‘R is X mod Y.‘
17. **FOL to CNF conversion Step 1** → Eliminate implications/biconditionals ($A \rightarrow B \Rightarrow \neg A \vee B$).
18. **Move negations inward FOL rule** → $\neg\forall x P(x) \Longrightarrow \exists x \neg P(x)$.
19. **Standardize variables step** → Renaming variables to ensure unique names per quantifier.
20. **Standard CNF format** → Conjunction of disjunction of literals (conjunction of clauses).
21. **Resolution Refutation proof method** → Negate target goal, resolve with KB clauses to derive $\square$.
22. **Resolvent of $P(x) \vee Q(x)$ and $\neg P(A)$ using $\{x/A\}$** → $Q(A)$.

23. **Semantic Network representation** → Concept nodes connected by directed, labeled relation edges.
24. **Frame-based representation attributes and values** → Slots (attributes) and fillers (values).
25. **Inheritance in Frames** → Subclasses inherit default slot values from superclasses.
26. **Fuzzy Set membership values** → Real numbers representing membership degree in $[0, 1]$.
27. **Fuzzy Union operator ($\cup$)** → $\mu_{A\cup B}(x) = \max(\mu_A(x), \mu_B(x))$.
28. **Fuzzy Intersection operator ($\cap$)** → $\mu_{A\cap B}(x) = \min(\mu_A(x), \mu_B(x))$.
29. **Fuzzy Complement operator ($\bar{A}$)** → $\mu_{\bar{A}}(x) = 1 - \mu_A(x)$.
30. **Difference between Crisp and Fuzzy Set** → Crisp has membership $\{0, 1\}$; Fuzzy has degree $[0, 1]$.
31. **Hebb's Rule neuroscience logic** → Neurons firing together wire together ($\Delta w_{ij} = \eta a_i a_j$).
32. **Crossover operator in Genetic Algorithms** → Swapping genetic segments between two parent chromosomes.
33. **Mutation operator in Genetic Algorithms** → Randomly flipping gene values to maintain diversity.